



Quantum anomalous Hall states in Li/Na-doped kagome V_3X_8 ($X = \text{Cl}, \text{Br}$) monolayers

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ARTICLE INFO

Keywords:

Quantum anomalous Hall insulator
Ferromagnetic
Kagome lattice
Out-of-plane magnetic anisotropy

ABSTRACT

The quantum anomalous Hall (QAH) insulators have great potential in low-energy electronic devices due to their dissipationless conducting channels. By carrying first-principles calculations, we predict that Li/Na-doped two-dimensional kagome lattice V_3X_8 ($X = \text{Cl}, \text{Br}$) monolayers, $V_3X_8@Li(Na)$, are intrinsically ferromagnetic (FM) half-semimetals with out-of-plane magnetic anisotropy. Interestingly, the $V_3X_8@Li(Na)$ monolayers exhibit QAH effect with Chern number of $C = 1$, and a topological phase transition from $C = 1$ to $C = -1$ occurs when rotating the magnetization direction from z axis to $-z$ axis. Moreover, the magnetic and topological properties of $V_3X_8@Li(Na)$ monolayers can be further modulated by biaxial strains, and FM QAH insulator and antiferromagnetic (AFM) semiconductor transition can be emerged. Our work provides that $V_3X_8@Li(Na)$ monolayers can serve as materials for exploring novel QAH physics.

1. Introduction

The quantum anomalous Hall (QAH) effect is a new class of topological phenomena for fundamental condensed matter physics and material science [1]. Contrary to the quantum Hall effect, the QAH effect originates from intrinsic magnetic moments and spin-orbit coupling (SOC) [2–6]. Two-dimensional (2D) QAH insulator, which is featured by transverse Hall conductance quantized to an integer C times e^2/h at zero external magnetic field (C is called Chern number), has edge states carrying electron currents which cannot be dissipated due to the absence of backscattering [7]. Therefore, realization of the QAH insulator and the utilization of its chiral edge states is advantage to the design of new low-energy consumption quantum electronic devices.

Up to now, various 2D QAH candidates were experimentally and theoretically proposed based on different mechanisms. For example, the QAH effect has been observed in transition metal atom (TM) doped topological insulators like $\text{Bi}(\text{Sb})_2\text{Te}_3$ [7], odd-layered MnBi_2Te_4 [8,9], as well as twisted bilayer graphene [10,11], etc. Besides, a few 2D intrinsic ferromagnetic (FM) monolayers, such as transition metal (TM) doped quantum spin Hall insulators [12,13], 2D metal-organic frameworks [14,15], 2D TM halides [16–21], 2D TM oxides [22,23], 2D metal nitride [24], 2D TM chalcogenides [25–30], etc., have also been

predicted to hold QAH effect. Recently, it has been discovered that the QAH effect can also be achieved in antiferromagnetic (AFM) systems, such as CrO [31], MnO [32], RbCr_2S_4 [33], bilayer Mn_3Sn [33], NiRuCl_6 [34], and so on. All the theoretical and experimental discoveries boost the development of QAH effect in real materials.

The advent of kagome lattice, which is composed of corner-sharing triangles geometry, provides a significant platform to study QAH effect due to its unique crystal structure analogous to the honeycomb lattice [35,36]. Muechler et al. predicted that $\text{Co}_3\text{Sn}_3\text{S}_2$ monolayer is a QAH insulator with $C = 3$ [37]. Xing et al. revealed that V_3F_8 monolayer can be QAH insulator with a topological flat band near Fermi level under uniaxial strain [38]. Very recently, TM_3X_4 monolayers are identified to be a stable $C = 1$ QAH insulator [26,30]. Nevertheless, the number of 2D candidates with kagome lattice reported are still very limited, therefore, searching more 2D layered kagome materials with intrinsic QAH properties remain an important frontier for discovery.

In this work, by using first-principles calculations, we systematically investigated the structural, magnetic, electronic, and topological properties of Li/Na-doped kagome lattice V_3X_8 ($X = \text{Cl}, \text{Br}$) monolayers, $V_3X_8@Li(Na)$, which may be synthesized by doping Li/Na atoms in the 2D kagome lattice TM_3X_8 in experiment. And to maintain the structural symmetry, we doped one Li/Na atom into the unit cell of V_3X_8

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<https://doi.org/10.1016/j.jalcom.2025.179103>

Received 7 November 2024; Received in revised form 3 February 2025; Accepted 8 February 2025

Available online 9 February 2025

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monolayers. The results show that all the doped structures are FM half-semimetals with 100 % spin-polarized currents and out-of-plane magnetic anisotropy. Furthermore, $V_3X_8@Li(Na)$ exhibit QAH effect with $C = 1$, and a topological phase transition from $C = 1$ to $C = -1$ can be realized when the magnetization direction rotating from z axis to $-z$ axis. In addition, the FM QAH insulator -AFM semiconductor transition can be occurred in $V_3X_8@Li(Na)$ under biaxial tensile/compressive strains.

2. Computational methods

Our calculations based on density functional theory (DFT) are performed in the Vienna *ab initio* Simulation Package (VASP) [39,40]. The exchange-correlation interaction is described by a general gradient approximation of the Perdew-Burke-Ernzerhof (PBE) function [41]. The projector-augmented wave potential (PAW) method [42] is used to describe the interaction between valence electrons and ion nuclei by setting the cut-off energy of the plane wave basis of 500 eV. The vacuum spaces are set to be $> 15 \text{ \AA}$ and the GGA+ U method with $U_{eff} = 3.0 \text{ eV}$ is used for V- d electrons based on previous studies [43]. Geometry optimization and electronic structure calculations are performed using $9 \times 9 \times 1$ and $27 \times 27 \times 1$ k -meshes, respectively. The convergence criteria for energy and force are 10^{-6} eV and $0.01 \text{ eV/\AA}$, respectively. On the basis of DFPT [44], phonon dispersions are calculated using $2 \times 2 \times 1$ supercells by the Phonopy code [45] to identify the dynamic stability, with the convergence thresholds of 10^{-8} eV and $10^{-4} \text{ eV/\AA}$ for energy and force, respectively. The *ab initio* molecular dynamics (AIMD) simulations [46] with $2 \times 2 \times 1$ supercells at the temperatures of 300 K using a Nose-Hoover thermostat method [47] in the NVT ensemble. The Curie temperatures (T_C s) of $V_3X_8@Li(Na)$ monolayers are calculated using the EspinS package [48], in which 20×20 supercells are adopted in the Monte Carlo (MC) simulations. The edge states are calculated by using the WannierTools package [49].

3. Results and discussions

3.1. Structure and stability

The top and side views of the Li/Na-doped monolayer V_3X_8 ($X = Cl, Br$), $V_3X_8@Li(Na)$, are shown in Fig. 1a, in which the V atoms form a 2D kagome lattice and each V atom is surrounded by six nearest-neighboring X atoms forming an edge-sharing distorted octahedral. The space group of $V_3X_8@Li(Na)$ is $P\bar{3}m1$ (no. 164), in which one Li/Na atom, three V atoms and eight X atoms are included in one unit cell, as shown in the dashed line in Fig. 1a. The optimized lattice constants (a_0) of $V_3X_8@Li(Na)$ are in the range of $7.20 \text{ \AA}$ - $7.78 \text{ \AA}$, which are slightly larger than that of V_3Cl_8 monolayer ($7.16 \text{ \AA}$) [50] and V_3Br_8 monolayer ($7.55 \text{ \AA}$) [50], the bond lengths between V and neighbouring X atoms are around $2.43 \text{ \AA}$ - $2.67 \text{ \AA}$. Detailed geometric information are list in Table S1 and Fig. S1 in Supporting Information. The electron localization function (ELF) of V_3Cl_8 and $V_3Cl_8@Li$ monolayers are shown in Fig. 1e and f, it is found that after Li doping, the electrons around X atoms are more localized, indicating that the stronger ionic characteristic of V-X and Li-X bonds after Li doping.

To examine the dynamic stability of the $V_3X_8@Li(Na)$ monolayers, the phonon dispersion were calculated. As shown in Fig. 1b and Fig. S2, the frequencies of $V_3X_8@Li(Na)$ monolayers within the Brillouin zone (BZ) are all positive, implying that they are dynamically stable. To further evaluate the thermodynamic stabilities of these $V_3X_8@Li(Na)$ monolayers, the AIMD simulations at 300 K are calculated. The free energies variation during AIMD simulation after 6ps at 300 K are shown in Fig. S3 and the final structures at 6ps are also plotted. Clearly, all the $V_3X_8@Li(Na)$ monolayers maintain their structures with slight deformation, suggesting their thermal stability. Meanwhile, the formation energy (E_f) is calculated with the definition of

$$E_f = \left[3 * \mu_V + 8 * \mu_{Cl/Br} + \mu_{Li/Na} - E_{V_3X_8@Li(Na)} \right] / 12 \quad (1)$$

where $E_{V_3X_8@Li(Na)}$ is the total energy of $V_3X_8@Li(Na)$ monolayers, μ_V and $\mu_{Li/Na}$ are the chemical potential of V and Li/Na atoms in their bulk, respectively. $\mu_{Cl/Br}$ is the chemical potential of Cl/Br in Cl_2/Br_2

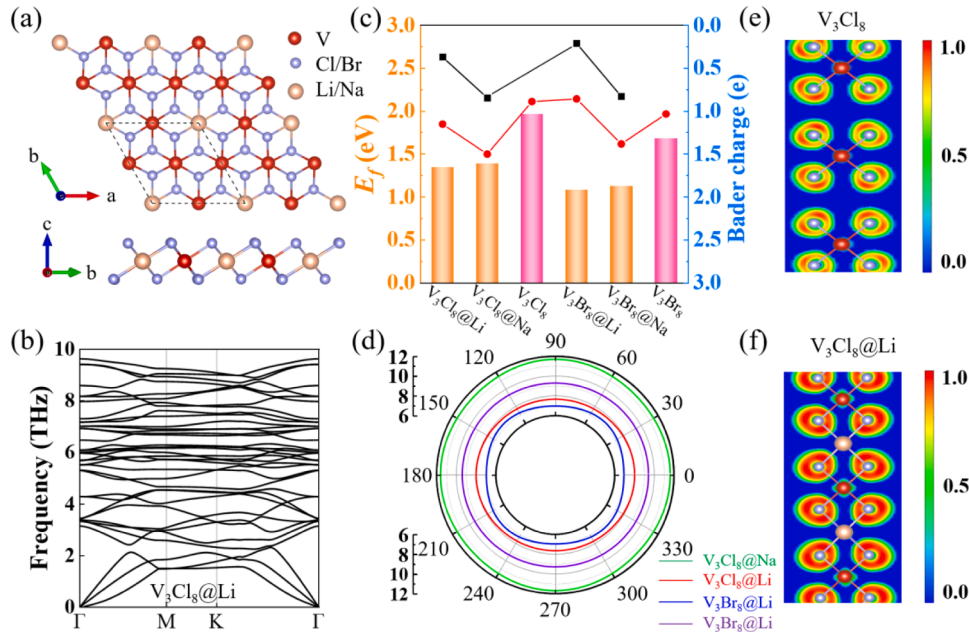


Fig. 1. (a) Top and side views of the $V_3X_8@Li(Na)$ monolayers. The dashed line denotes the unit cell. (b) Phonon spectrum of $V_3Cl_8@Li$ monolayer. (c) The E_f s (column) and Bader charge (symbol lines) of $V_3X_8@Li(Na)$ and V_3X_8 monolayer. Red and black lines represent the electrons depleted from V and Li/Na atoms, respectively. (d) Angular dependence of the Young's modulus $C^{2D}(\theta)$ for $V_3X_8@Li(Na)$ monolayers. The electron localization function (ELF) of V_3Cl_8 (e) and $V_3Cl_8@Li$ (f) monolayer.

molecules. As shown in orange columns in Fig. 1c, the E_F of $V_3X_8@Li$ (Na) monolayers are positive and around 1.08–1.39 eV per atom, which are slightly smaller than that of V_3X_8 monolayers (1.68–1.96 eV), indicating that these Li/Na doped systems are energetically favorable and feasible to be synthesized in experiment. Such stability of $V_3X_8@Li$ (Na) monolayers can be attributed to the electron transfer from metal atoms to X atoms, as indicated by the Bader charge analysis (symbol lines in Fig. 1c). In $V_3X_8@Li$ (Na) monolayers, X atoms gain less electrons from V atoms compared with that in V_3X_8 monolayers, but additional electrons can be transferred from Li/Na atoms to X atoms, and the electron depletion on Li atoms (0.37 e and 0.21 e for $V_3Cl_8@Li$ and $V_3Br_8@Li$) is less than that on Na atoms (0.84 e and 0.83 e for $V_3Cl_8@Na$ and $V_3Br_8@Na$), which is responsible for the more stable character of $V_3X_8@Na$ monolayers.

In addition, the mechanical stability of $V_3X_8@Li(Na)$ monolayers are accessed. The elastic tensor C can be expressed as:

$$C = \begin{pmatrix} C_{11} & C_{12} & 0 \\ C_{21} & C_{22} & 0 \\ 0 & 0 & C_{66} \end{pmatrix} \quad (2),$$

here, C_{ij} are the elastic constants. The calculated values of C_{11} , C_{22} , C_{12} and C_{66} (see [Table S2](#)) satisfy the Born criteria of mechanical stability with C_{11} , C_{22} , $C_{66} > 0$; $C_{11}C_{22} - C_{12}^2 > 0$. The in-plane Young's modulus $C^{2D}(\theta)$ and Poisson's ratio $\nu^{2D}(\theta)$ can be calculated by the following two formulas:

$$C^{2D}(\theta) = \frac{C_{11}C_{22} - C_{12}^2}{C_{22}\cos^4(\theta) + A\cos^2(\theta)\sin^2(\theta) + C_{11}\sin^4(\theta)} \quad (3),$$

$$\nu^{2D}(\theta) = \frac{C_{12}\cos^4(\theta) - B\cos^2(\theta)\sin^2(\theta) + C_{12}\sin^4(\theta)}{C_{22}\cos^4(\theta) + A\cos^2(\theta)\sin^2(\theta) + C_{11}\sin^4(\theta)} \quad (4)$$

where $A = (C_{11}C_{22} - C_{12}^2)/C_{66} - 2C_{12}$, $B = C_{11} + C_{22} - (C_{11}C_{22} - C_{12}^2)/C_{66}$; θ is the angle with x axis, $\theta = 0^\circ$ and $\theta = 90^\circ$ represent the x axis and y axis, respectively. The $C^{2D}(\theta)$ s as a function of θ are shown in Fig. 1d, it is found that in-plane Young's modulus of all the $V_3X_8@Li(Na)$ monolayers are small (less than 35 N/m) and anisotropic, which are much less than that in graphene (342.2 N/m) [51] and MoS_2 (120 N/m) [52], indicating that they have promising applications in flexible devices. The Poisson's ratios of $V_3X_8@Li(Na)$ monolayers are anisotropic and shown in Fig. S4, ranging from 0.29 to 0.38. It suggests that for $V_3Cl_8@Li(Na)$ monolayers, the Poisson's ratio has the largest values along the direction of $\theta = 45^\circ$, 135° , 225° , and 315° and has the smallest values along x/y directions. Contrary case is found for $V_3Br_8@Li(Na)$ monolayers, that is, the largest Poisson's ratios are found along the x/y directions, and the smallest Poisson's ratios are found along the $\theta = 45^\circ$, 135° , 225° , and 315° directions.

3.2. Magnetic properties

To determine the ground states of these $V_3X_8@Li(Na)$ monolayers, five different magnetic configurations are considered, including one FM state and four AFM states (see Fig. S5), and their relative energies are listed in Table S3. It is suggested that all the $V_3X_8@Li(Na)$ monolayers favor FM ground states, which are lower in energy than the lowest energy AFM states by about 53.30–233.22 meV per rectangular cell (see Fig. S4). The FM coupling of $V_3X_8@Li(Na)$ monolayers are arisen from the super-exchange interactions according to Goodenough-Kanamori-Anderson rule [53–55], as the bonds angles of $\angle V-X-V$ are close to 90° (see Table S1). The spin density distribution of FM $V_3Cl_8@Li$ monolayer is shown in Fig. 2a, it can be seen that the magnetic moments are mainly contributed by V atoms with the local magnetic moments of 2.5 μ_B , which is comparable to the V_3X_8 monolayers (7 μ_B /unit cell) [50]. The

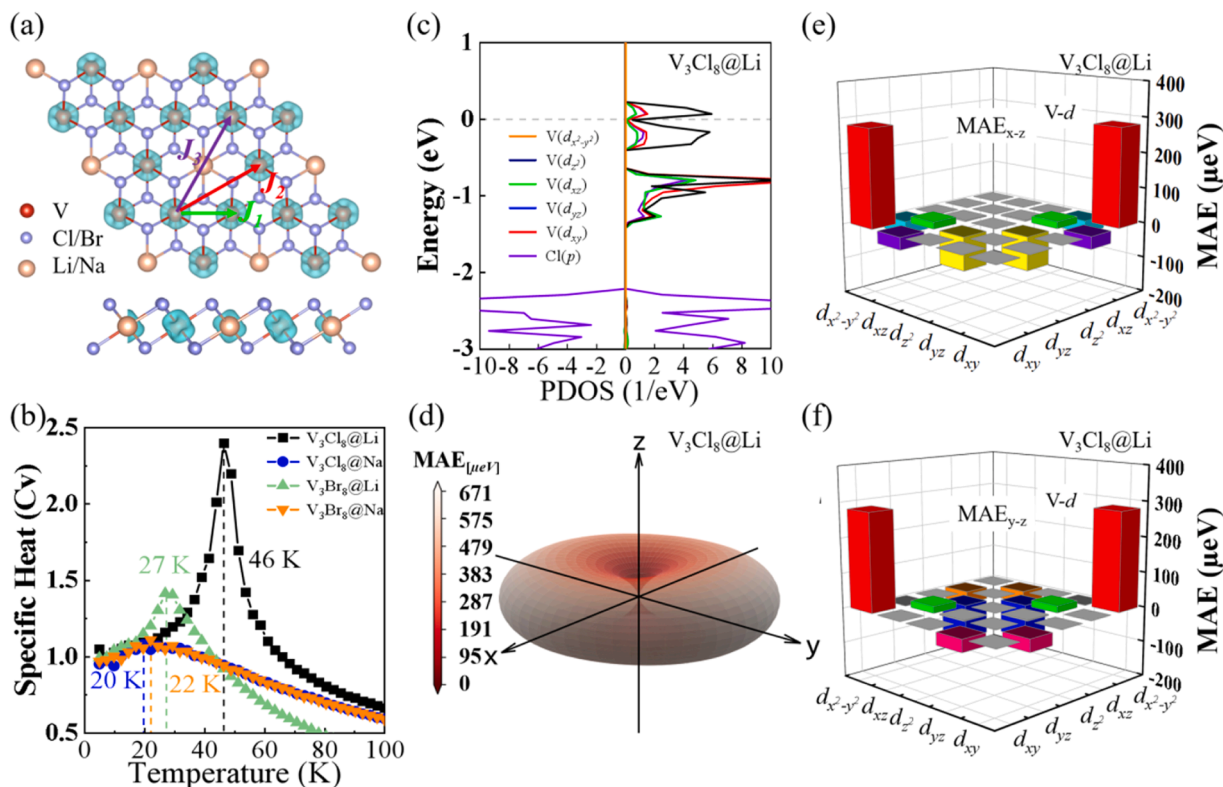


Fig. 2. (a) Spin density of $V_3Cl_8@Li$ monolayer. J_1 , J_2 and J_3 are the nearest, next-nearest (NN) and next-next-nearest (NNN) exchange parameters, respectively. (b) The specific heat as functions of temperature for $V_3X_8@Li(Na)$ monolayers. (c) PDOS of $V_3Cl_8@Li$ monolayer. (d) Angular dependence of MAE of $V_3Cl_8@Li$ monolayer with the magnetization direction in the whole space. The contribution of V-*d* orbital to MAE from the SOC interaction of $V_3Cl_8@Li$ monolayer along the *x* (e) and the *y* (f) directions. The energy is referenced to the *z* direction.

projected density of states (PDOS) plots (see Fig. 2c and Fig. S6) show that the magnetic moments of these systems are contributed by $V-d_z^2/d_{xz}/d_{xy}$ orbital electrons. In addition, the PDOS around the Fermi level are occupied by spin down electrons, leading to 100 % spin polarization ratio, which endows the $V_3X_8@Li(Na)$ monolayers with great advantages in spintronic applications.

The magnetic anisotropy energy (MAE) of $V_3X_8@Li(Na)$ monolayers in whole space are examined and shown in Fig. 2d and Fig. S7, which reveal that all $V_3X_8@Li(Na)$ monolayers tend to magnetize along the out-of-plane direction with the MAE values of 0.22, 0.09, 0.68 and 0.27 meV per V atom for $V_3Cl_8@Li$, $V_3Cl_8@Na$, $V_3Br_8@Li$, and $V_3Br_8@Na$, respectively. The MAE values are much larger than those of V_3F_8 (0.1 meV/V) [38], VCl_2 (0.25 meV/V) [56], Fe_2Cl (0.039 meV/Fe) [57], and comparable with those of $LiFeS$ (0.3 meV/Fe) [58] and $LiFeSe$ (0.2 meV/Fe) [58]. Moreover, the relative contributions of the spin-orbit coupling (SOC) interactions between different orbitals to the MAE are plotted in Fig. 2e, f and Figs. S8-S11, it is suggested that the magnetization is mainly contributed by the $V-(d_{xy}, d_{x^2-y^2})$ hybridization orbitals, while that from $Li(Na)$ atoms and $X-p$ orbitals are negligible. By performing MC simulations, we have estimated the T_C s of FM $V_3X_8@Li(Na)$ monolayers based on 2D Heisenberg Hamiltonian model, which is described by:

$$H = -J_1 \sum_{\langle ij \rangle} S_i \cdot S_j - J_2 \sum_{\langle\langle ij \rangle\rangle} S_i \cdot S_j - J_3 \sum_{\langle\langle\langle ij \rangle\rangle\rangle} S_i \cdot S_j \quad (5)$$

where S_i is the spin vector of site i , J_1 , J_2 and J_3 are the nearest (N), next-nearest (NN) and next-next-nearest (NNN) exchange coupling parameters between V atoms (see Fig. 2a). The calculated J_1 , J_2 and J_3 values and the calculation details are shown in Table S3 and Part I in SI. Such small J_i values indicate the FM spin interactions are weak, which lead to the T_C s of $V_3X_8@Li(Na)$ monolayers are low. The specific heat as a

function of temperature (see Fig. 2d) suggests that the T_C s are 46 K, 20 K, 27 K and 22 K for $V_3Cl_8@Li$, $V_3Cl_8@Na$, $V_3Br_8@Li$, $V_3Br_8@Na$, respectively, which are comparable to V_3F_8 monolayer (30 K) [38].

3.3. Electronic and topological properties

The band structures of $V_3X_8@Li(Na)$ monolayers are shown in Figs. 3 and 4a, e. Without considering SOC effect, all the $V_3X_8@Li(Na)$ monolayers are half-semimetals with 100 % spin polarization. In the spin-up channel, there are bands crossing the Fermi level at K-point with a twofold-degenerate Weyl point. Compared with the V_3X_8 monolayers, the doping of Li/Na atoms in the $V_3X_8@Li(Na)$ monolayers shifts the Fermi level in V_3X_8 , leading to the Weyl point riding on the Fermi level. While in the spin-down channel, large band gaps of 5.02 eV, 4.28 eV, 5.08 eV, and 4.36 eV are obtained for $V_3Cl_8@Li$, $V_3Br_8@Li$, $V_3Cl_8@Na$ and $V_3Br_8@Na$ monolayers, respectively.

Upon applying SOC effect with out-of-plane magnetization direction, the Weyl points of $V_3X_8@Li(Na)$ monolayers are opened, and the topological band gaps are 30.59 meV, 53.97 meV, 21.03 meV and 38.24 meV for $V_3Cl_8@Li$, $V_3Br_8@Li$, $V_3Cl_8@Na$ and $V_3Br_8@Na$ monolayers, respectively, as plotted in Figs. 3b, f and 4b, f. The electronic properties of $V_3X_8@Li$ ($X = Cl, Br$) monolayers are further confirmed by using the hybrid functional (HSE06) method, as shown in Fig. S12 in SI, which are similar with the results using PBE+U methods. When the magnetization is set along x/y directions, the Weyl points are maintained. Such results suggest that these $V_3X_8@Li(Na)$ monolayers maybe potential intrinsic QAH systems. Taking the $V_3Cl_8@Li$ monolayer as an example, we further analyze its topological property. As discussed above, the space group of $V_3X_8@Li(Na)$ monolayers is $P\bar{3}m1$ (No. 164), which can be characterized by a threefold rotation symmetry (C_{3z}), twofold rotation symmetry C_2 in-plane, mirror symmetry and inversion

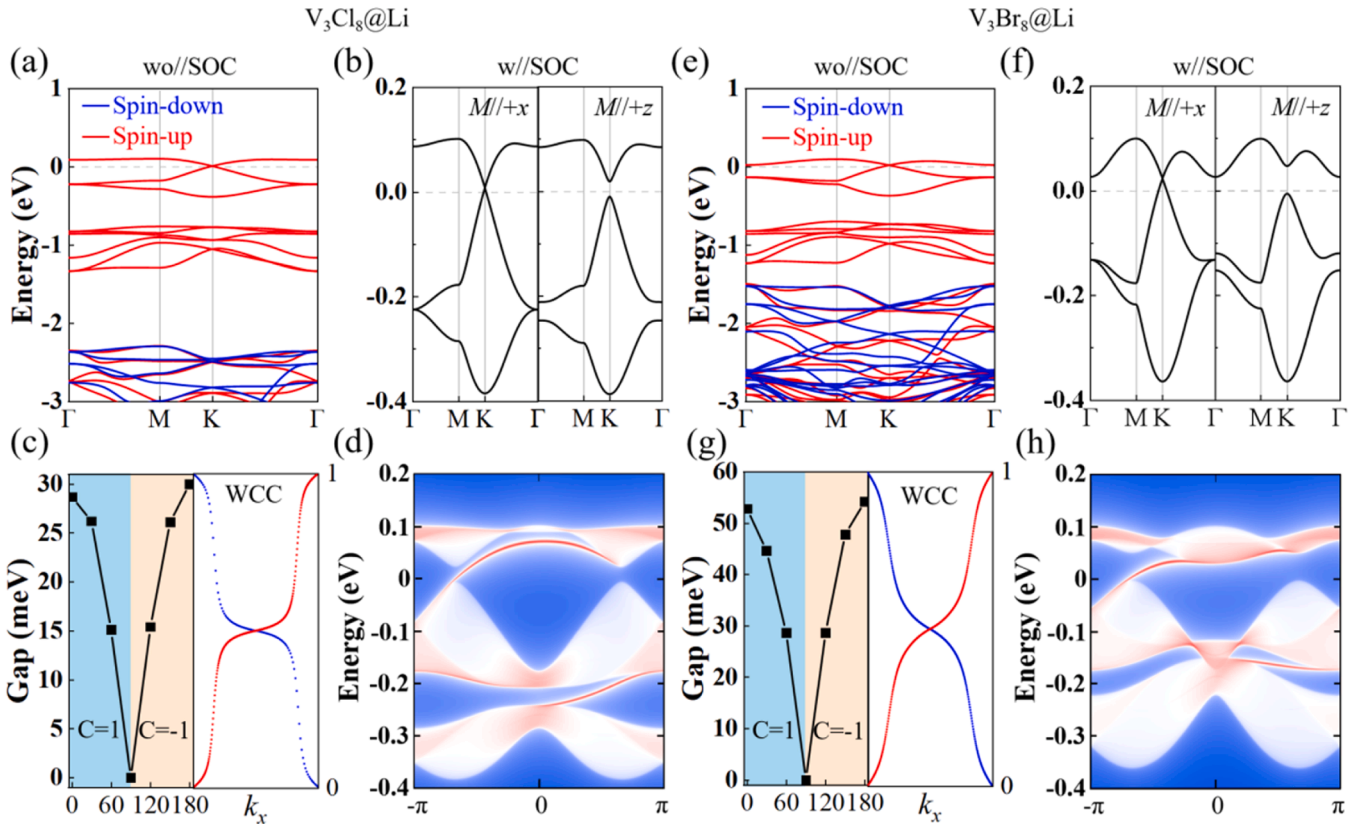


Fig. 3. Spin polarized band structures of $V_3Cl_8@Li$ and $V_3Br_8@Li$ monolayers without SOC (a, e) and with SOC when magnetization is along x and z direction (b, f). (d, h) Edge states of $V_3Cl_8@Li$ and $V_3Br_8@Li$ monolayers. The gap at K point, Chern number and WCC (the red line corresponds to $C = 1$ and the blue line corresponds to $C = -1$) of $V_3Cl_8@Li$ (c) and $V_3Br_8@Li$ (g) monolayers when the magnetization rotates from 0° to 180° in zy plane.

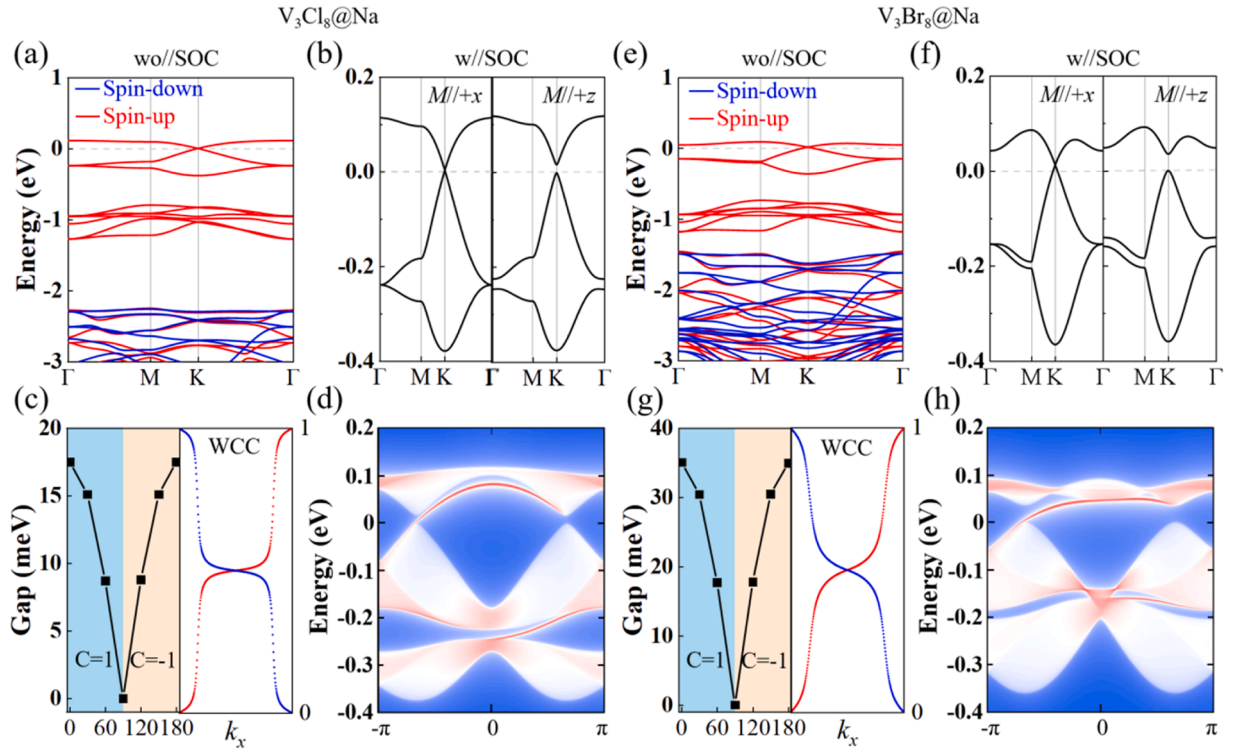


Fig. 4. Spin polarized band structures of $V_3Cl_8@Na$ and $V_3Br_8@Na$ monolayers without SOC (a, e) and with SOC (b, f) when magnetization is along x and z direction. (d, h) Edge states of $V_3Cl_8@Li$ and $V_3Br_8@Li$ monolayers. The gap at K point, Chern number and WCC (the red line corresponds to $C = 1$ and the blue line corresponds to $C = -1$) of $V_3Cl_8@Li$ (c) and $V_3Br_8@Li$ (g) monolayers when the magnetization rotates from 0° to 180° in zy plane.

symmetry P (see Fig. 5a). As shown in Fig. 5b, when the magnetization is along x direction ($\theta = 0^\circ$), both the C_{3z} and C_2 symmetries are broken, while the M_x symmetry is retained. As a result, the Weyl point is preserved, and the systems behave as a Weyl semimetal with two Weyl points manifested in the edge states, which are connected by a Fermi arc

(see Fig. 5c). When the magnetization is along the $\theta = 30^\circ$ direction (see Fig. 5d, e), both the C_{3z} and all mirror symmetries are broken, while the C_{2i} symmetry remains. In this case, the Weyl point is still kept, and the system exhibits Weyl semimetal characteristics. Therefore, only those flux configurations that simultaneously break both mirror symmetry and

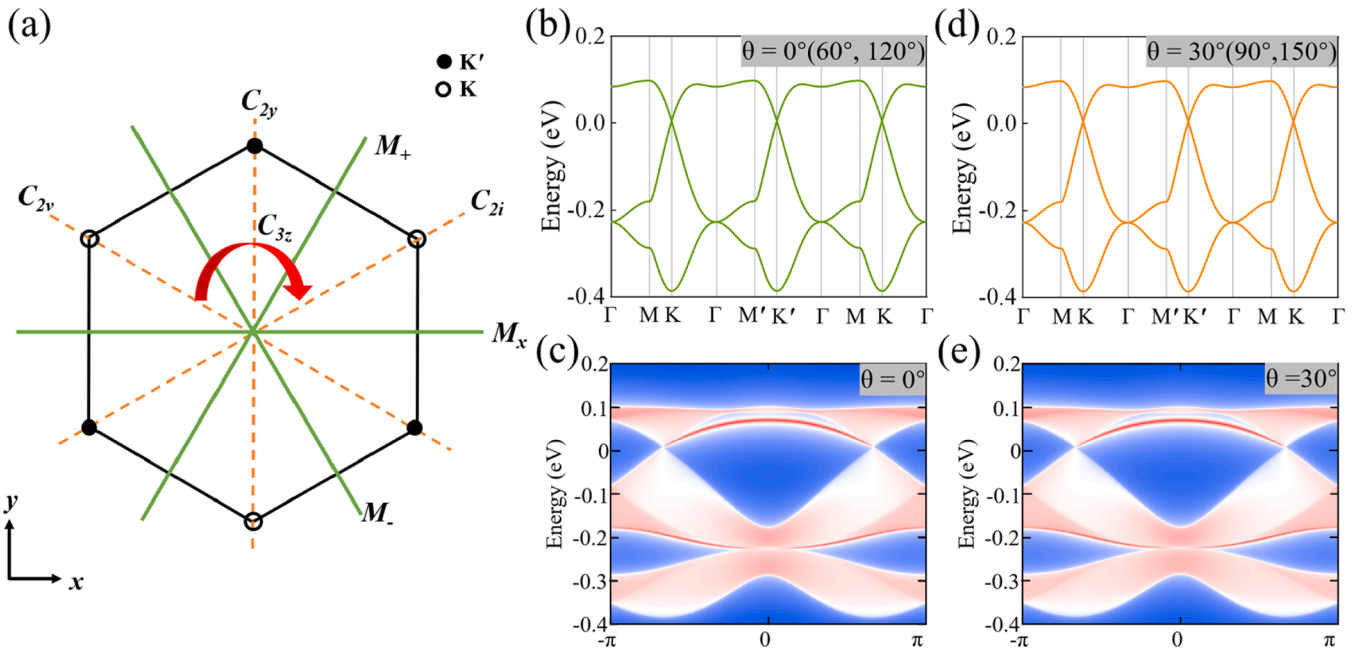


Fig. 5. (a) Brillouin zone of $V_3X_8@Li(Na)$ monolayers. Red arrow indicates the C_{3z} symmetry, orange and green lines represent the C_2 and mirror symmetries, respectively. Global band structures with SOC of the $V_3X_8@Li(Na)$ monolayers for (b) $\theta = 0^\circ$ ($60^\circ, 120^\circ$) and (d) $\theta = 30^\circ$ ($90^\circ, 150^\circ$). The edge state in the presence of SOC when magnetization is aligned along +z axis of the $V_3Cl_8@Li$ monolayer for angles (c) $\theta = 0^\circ$ and (e) $\theta = 30^\circ$. The angle θ indicates the orientation of the magnetization relative to the x-axis.

C_2 symmetry of $V_3X_8@Li(Na)$ monolayers can create gaps at the high symmetry point K. As shown in Figs. 3 and 4 and Figs. S13-S16, by introducing an out-of-plane magnetism, the Weyl point of the system is gaped, resulting in the formation of a topological insulator. To determine the topological properties of $V_3X_8@Li(Na)$ monolayers, we determine their Chern numbers according to Eq. (10) as below:

$$C = \frac{1}{2\pi} \int_{BZ} d^2k \Omega(k) \quad (10)$$

where $\Omega(k)$ is the Berry curvature integrating over the BZ. The $V_3X_8@Li(Na)$ monolayers are proved to be $C = 1$ QAH insulators, the topological identification of edge states projected on the (010) surface are plotted in Figs. 3 and 4d, h.

The band structures of $V_3X_8@Li(Na)$ monolayers considering SOC effect with magnetization direction in zy and xy planes are shown in Figs. 13-S16. For $V_3Cl_8@Li$, $V_3Br_8@Li$, $V_3Cl_8@Na$ and $V_3Br_8@Na$ monolayers, the topological non-trivial gap at K point gradually

decreases from 30.59, 53.97, 21.03 and 38.24 meV with the magnetization direction varying from 0° to 90° and reopens with the magnetization direction from 90° to 180° . It means that $V_3X_8@Li(Na)$ monolayers occur a topological phase transition from Chern number $C = 1$ to $C = -1$, as confirmed by the wannier charge center (WCC) in Figs. 3 and 4c, g. On the contrary, when magnetization direction rotates in the xy planes, the Weyl point at the K point is preserved, maintaining the half-semimetallic properties.

3.4. Strain effect

Finally, the strain effect on the electronic and topological properties of $V_3X_8@Li(Na)$ monolayers are investigated. The strain intensity, ε , is the ratio between the lattice constant change (Δa) under biaxial strain and a_0 , $\varepsilon = \Delta a/a_0$, the positive and negative values represent biaxial tensile and compressive strains, respectively. Here, the ε is in the range of $[-8\%, 8\%]$. The energy difference between FM and the most stable

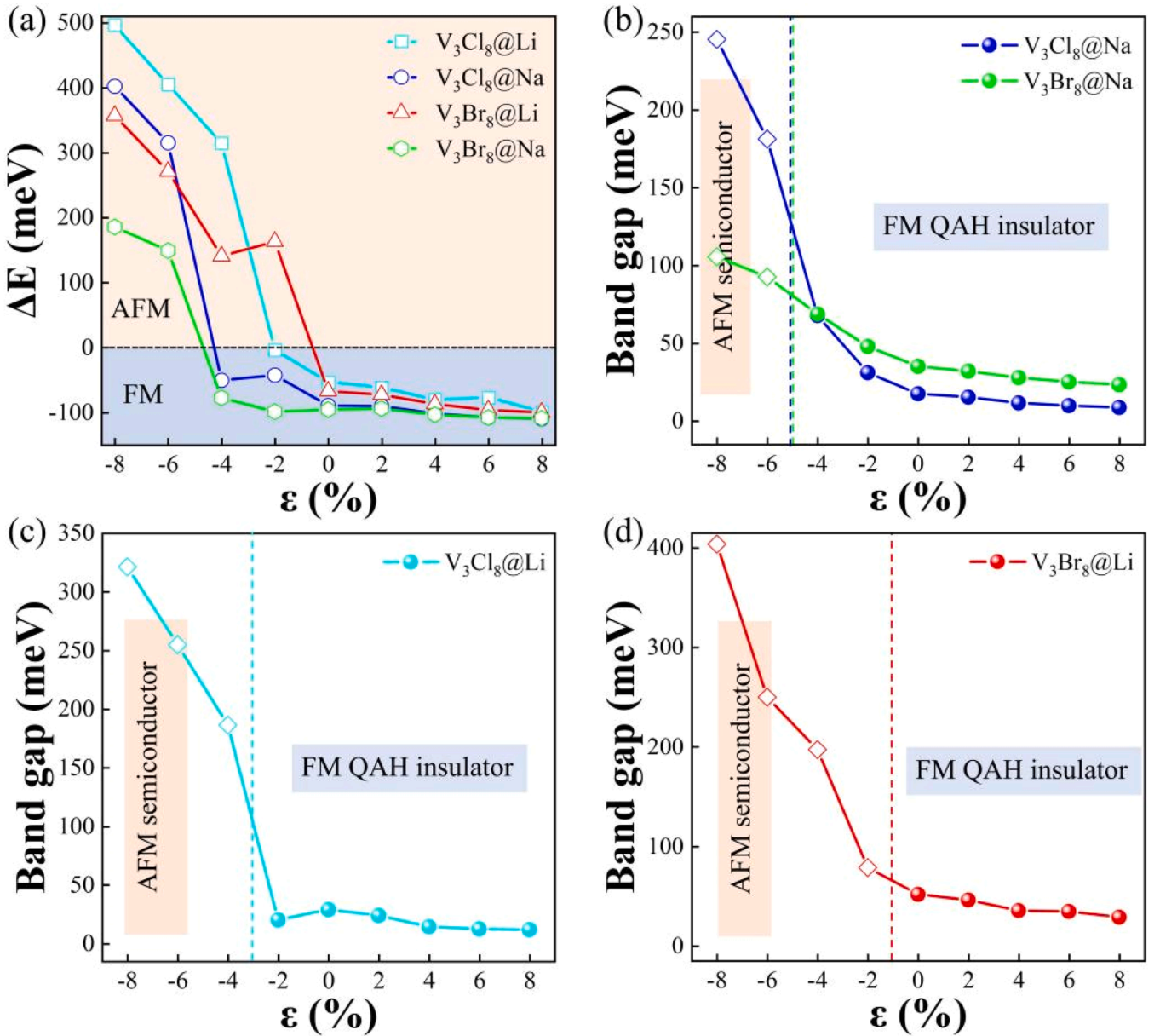


Fig. 6. (a) The energy difference between FM and most stable AFM state of $V_3X_8@Li(Na)$ monolayers under biaxial strains with the ε range of $[-8\%, 8\%]$. (b–d) The band gaps of AFM $V_3X_8@Li(Na)$ monolayers (the rhombus symbol) and the topological non-trivial gaps at K point of FM $V_3X_8@Li(Na)$ monolayers (the sphere symbol).

AFM state, $\Delta E = E_{\text{FM}} - E_{\text{AFM}}$, is shown in Fig. 6a. Clearly, all the $\text{V}_3\text{X}_8\text{@Li}$ (Na) monolayers keep their FM ground states under biaxial tensile strains till 8 %, while FM-AFM transition occurs when the applied biaxial compressive strains to -2 % ($\text{V}_3\text{Br}_8\text{@Li}$), -4 % ($\text{V}_3\text{Cl}_8\text{@Li}$) and -6 % ($\text{V}_3\text{Cl}(\text{Br})_8\text{@Na}$), respectively. The band structures of these $\text{V}_3\text{X}_8\text{@Li}(\text{Na})$ monolayers under different biaxial strains are shown in Figs. S17–S20. The band structures and edge states of $\text{V}_3\text{X}_8\text{@Li}(\text{Na})$ monolayers under $\varepsilon = 4\%$ biaxial tensile strain with magnetization along the z direction are shown in Fig. S21, demonstrating that the ferromagnetic $\text{V}_3\text{X}_8\text{@Li}(\text{Na})$ monolayers under biaxial strains are robustly topological. While the AFM ones are changed to be semi-conducting properties. The band gaps for AFM semiconductors and the topological non-trivial gaps at K point for FM QAH insulators are plotted in Fig. 6b–d. It is shown that the band gaps of AFM monolayers under compressive strains increases steeply with the increase of the compressive strains. In contrast, the topological non-trivial gaps of the FM candidates under tensile strains decrease gradually with the increase of strains, which the topological non-trivial gaps are in the range of [11.83 meV, 28.96 meV], [28.68 meV, 51.62 meV], [8.6 meV, 67.7 meV] and [23.2 meV, 68.68 meV], for $\text{V}_3\text{Cl}_8\text{@Li}$, $\text{V}_3\text{Br}_8\text{@Li}$, $\text{V}_3\text{Cl}_8\text{@Na}$ and $\text{V}_3\text{Br}_8\text{@Na}$ monolayers, respectively.

As shown in Fig. S22, the easy magnetization axes of $\text{V}_3\text{X}_8\text{@Li}(\text{Na})$ monolayers vary with the external strains. Under biaxial tensile strains, the out-of-plane easy magnetization axes are kept at small ε , which the absolute value of the MAE decreases with increasing ε . Besides, the out-of-plane to in-plane easy magnetization axes occur at $\varepsilon = 6\%$, 6% , 8% and 6% for $\text{V}_3\text{Cl}_8\text{@Li}$, $\text{V}_3\text{Cl}_8\text{@Na}$, $\text{V}_3\text{Br}_8\text{@Li}$ and $\text{V}_3\text{Br}_8\text{@Na}$, respectively. Under biaxial compressive strains, the easy magnetization axes of $\text{V}_3\text{Cl}_8\text{@Li}(\text{Na})$ monolayers remain out-of-plane, while for $\text{V}_3\text{Br}_8\text{@Li}(\text{Na})$ monolayers, the easy magnetization axes are changed to in-plane under $\varepsilon = -2\%$ (-6%).

4. Conclusions

In summary, based on first principles calculations, we predict that 2D kagome Li/Na-doped V_3X_8 ($\text{X} = \text{Cl}, \text{Br}$) monolayers, $\text{V}_3\text{X}_8\text{@Li}(\text{Na})$, are FM stable half-semimetals with 100 % spin polarization. Moreover, the $\text{V}_3\text{X}_8\text{@Li}(\text{Na})$ monolayers exhibit QAH effect with Chern number $C = 1$, and undergoes a topological phase transition from $C = 1$ to $C = -1$ when the magnetization direction rotates from z axis to $-z$ axis. Furthermore, all the $\text{V}_3\text{X}_8\text{@Li}(\text{Na})$ monolayers maintain the FM QAH effect under biaxial tensile strains within 8 %, and FM-AFM phase transition occurs when the biaxial compressive strain up to -2 %, -4 % and -6 % for $\text{V}_3\text{Br}_8\text{@Li}$, $\text{V}_3\text{Cl}_8\text{@Li}$ and $\text{V}_3\text{Cl}(\text{Br})_8\text{@Na}$, respectively. Our work provides a series of potential QAH candidates for practical electronic devices.

CRedit authorship contribution statement

Lu Jinlian: Validation. **Yao Xiaojing:** Writing – review & editing, Supervision. **He Ailei:** Supervision. **Zhang Xiuyun:** Writing – review & editing, Supervision, Conceptualization. **Lian Huijie:** Investigation, Formal analysis, Data curation. **Xu Xiaokang:** Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work is supported by Science and Technology Project of Hebei Education Department (BJK2024158), S&T Program of Hebei (22567617 H). We thank the computational resources at Hebei Normal University and Yangzhou University.

Conflicts of interest

There are no conflicts to declare.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jallcom.2025.179103.

Data availability

Data will be made available on request.

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